Roll No. :

END SEMESTER EXAMINATION DEC 2025

Name of the Course: M. Tech. (CSE)
Name of the Paper: Networking Protocols
Time: 3 Hours

Semester: I
Paper Code: MCS-104
Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer ***any two sub questions*** among a, b & c in each main question
(iii) Total marks for each main question are 20.
(iv) Each sub-question carries 10 marks

Q1	(20 marks)	
(a)	(i) Differentiate between circuit switched and packet switched network and identify advantages and limitations of each. (ii) At which layer a network switch normally operate? Which address does it use to facilitate communication? Identify and elaborate the requirements for Layer 3 switch in LAN.	CO1
(b)	Prepare a list of components required for a main network switch rack at a central control centre of an organization having computers in multiple buildings located in a campus of 3 KM^2. This organization needs Internet connectivity from two ISPs. List out the various components under the active, passive and other categories you need in the rack and their purpose, including security requirement.	
(c)	Consider a Token Ring (IEEE 802.5), with the help of a neat diagram, elaborate on collision-free communication in a LAN using details like architecture, media etc. Further, highlight differences in topology, access control, performance, and reliability between Token Ring and Ethernet.	
Q2	(20 marks)	
(a)	(i) Elaborate on need for controlled allocation of IP address. Which organizations controls the allocation globally, and regionally? (ii) What do you think is the requirement to classify Public and Private IP addresses? What are its advantages and restrictions of private addresses? Provide the range of private IP addresses in class A and B.	CO2
(b)	(i) Discuss the concept of Proxy ARP on allowing two separate physical networks with single IP Prefix. (ii) Consider a host H1 (192.168.50.10) on LAN 1 wants to send an IP packet to H3 (192.168.50.130) on LAN 2 (hidden). • Explain step-by-step how ARP resolution happens in this setup when Proxy ARP is enabled on the router. • Show the ARP request and ARP reply at a conceptual level (who sends what to whom).	
(c)	(i) Identify the limitations of RARP that motivated the development of Bootstrap Protocol. (ii) Consider a machine M1 on LAN1 makes use of BOOTP to obtain IP configuration, however the BOOTP server is on LAN2. What concept of BOOTP can be used in such cases to provide required config to M1? Explain with the help of neat figure depicting a typical network.	
Q3	(20 marks)	
(a)	(i) List out the problems/errors that data delivery (packets) may suffer during transmission from sender to destination over the Internet. Which protocol is used to convey the errors to sender? (ii)Provide the header format and value associated with various fields for: • Testing Destination Reachability and Status (Ping) • Detecting Circular Or Excessively Long Routes	CO3

(b)	(i) Consider designing a network using subnets for a multi-department organization. Elaborate on four key questions that must be answered to arrive at understanding the basis of design? (ii) An organization has been assigned a classful network number 129.35.0.0 and it needs to create a set of subnets that supports up to 58 hosts on each subnet. Considering the future requirements (provide binary and decimal representations): • Define extended prefix and subnet mask • Provide IP address for first, second, third and last subnet. • For each of the above, provide first and last valid host address	
(c)	(i) Provide the concept of Base header and extension header in IPv6 with schematic figures showing examples of: no extension header, one extension header and two extension headers. (ii)With reference to IPv6, explain the concept of Unicast, Anycast and Multicast with typical network diagrams.	
Q4	(20 marks)	
(a)	Provide a brief overview of communication using VPN over Internet. Consider an organization requiring communication between a corporate office in Delhi and branch office in Mumbai. Propose a schematic diagram showing establishment of VPN using various devices/components between the two offices and explain.	
(b)	(i) How does a NAT help in conserving IP addresses? Where would you place the NAT in a network? (ii) Consider a Server S1 providing three services. S1 is accessed by multiple clients and thus has to distribute its processing load to internal servers for satisfactory performance. Propose a solution using NAT depicting a network diagram with NAT server and NAT Table, internal servers, external client and communication packets with IP addresses. Provide proper explaination.	CO4
(c)	(i) Explain the concept of Private and Hybrid Networks with suitable architectural diagrams. (ii) Discuss why organizations would make use of both private and public networks, evaluating the security, scalability, and performance implications of hybrid networking.	
Q5	(20 marks)	
(a)	(i) Consider digitization of the voice signal using pulse code modulation (PCM) using two sampling frequencies of 6 KHz and 10 KHz. For *each of these:* • What is the sampling interval? • How many samples are generated per second? • Assuming each sample is represented using 16 bits. How much data is generated per second? • Discuss the ways to reduce the size of generated data and its trade-offs (ii) Evaluate the major challenges of VoIP such as delay, jitter, and packet loss, and suggest suitable mitigation techniques.	CO5
(b)	Analyse how the Domain Name System (DNS) translates human-readable domain names into IP addresses. Illustrate your answer with a neat, labelled diagram showing the sequence of communication between the initiator, local server/resolver, root server, TLD server, and authoritative name server.	
(c)	(i) What are labels in domain name? What are the restrictions/rules on assigning labels? Also provide at least four top level domains maintained by global authority and mention for which type of organizations are they allocated? (ii) VoIP calls are more sensitive to delay and jitter than live broadcasts. Why is this so? Further, are sequence numbers really necessary in RTP, or can only the timestamp be used instead? Explain	

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